

Molecular structure inference from molecular spectra

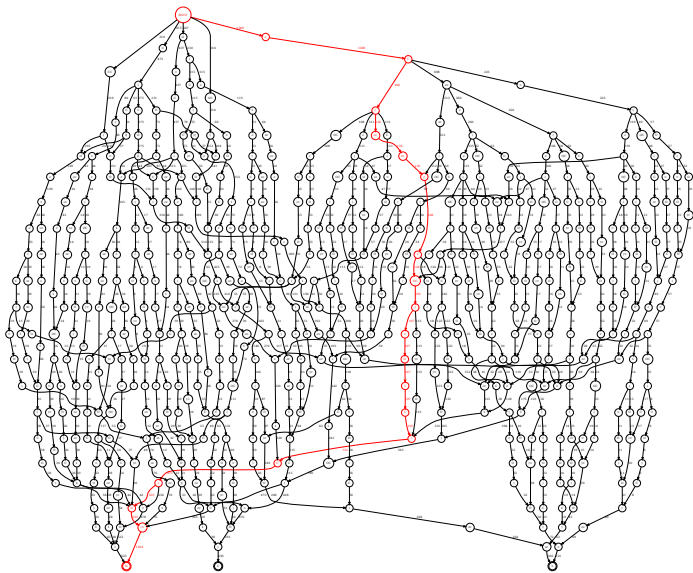
with

actual compact representations of names, yet mostly silly

Clustered halo-/endo- ensemble of sample embeddings
CHEESE

- PubChem SMILES “chembedded” with ChemBERTa-2 (384-d)
- Clustered in 1024 clusters by cosine distance
- Take another dataset D , e.g., Alberts et al. [1]
- Chembed D , sort into the clusters
- The outer 10% (halo) define the holdout set, plus
- same number of samples from the inner 90% (endo)
- Rest from the inner 90% (endo) is the training set

Graph representation integrating multiple alternate chemical equivalents
GRIMACE



Multi-task equilibrated normed training objective regression
MENTOR

Losses:

- $\frac{1}{n}(\frac{1}{2}\|T\mathbf{e}\|_2^2 - \log \det T + \frac{n}{2} \log 2\pi)$ – embedding ($n = 384$)
- $\frac{1}{m}(\|S\mathbf{e}\|_1 - \log \det S + m \log 2)$ – functional groups ($m = 289$)
- KL divergence, position-averaged – next-token distribution
- Causal LM = $-\log p(\text{eos} \mid \text{prefix})$ – end-of-sequence token

with trainable matrices T and S , combined à la Cipolla et al. [2]:

$$\text{minimize} \quad \sum_t \left(\frac{1}{\sigma_t^2} \text{softplus}(\text{loss}_t) + \log \sigma_t^2 \right)$$

with a trainable σ_t for each task t .

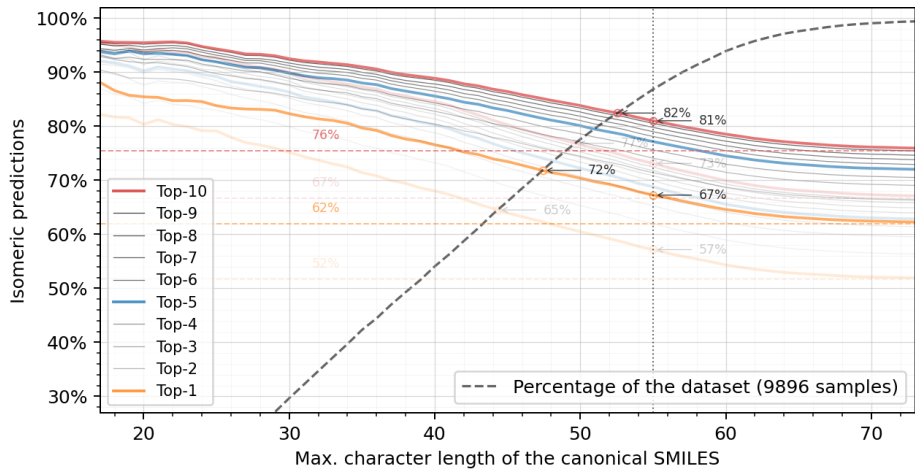
NMR annotation to multitask estimation
& learning equivalent SMILES sequences

Training setup:

NMR annotation $\rightarrow$      GRIMACE 

where

- , ,  are task-specific trigger tokens,
-  is where the hidden state is supervised,
- GRIMACE is the next-token distribution supervised SMILES,
-  is the supervised end-of-sequence token.



Selective heteromodal input fine-tuning SHIFT

Fine-tuning on selected modalities in equal measure:

- F+H
- F+H+Q
- F+H+C
- F+H+C+Q

~>

F+H+HSQC more informative (62% acc.) than F+H+C (59% acc.)

Data-informed confidence estimators

DICE



$0 \leq \text{is_exact} \leq 1$



$0 \leq \text{is_isomer} \leq 1$



$-0.32 \leq \text{score} \leq -0.13$



$0 \leq \text{is_valid} \leq 1$



$1 \leq \text{freq} \leq 10$



$0 \leq \text{is_majority} \leq 1$



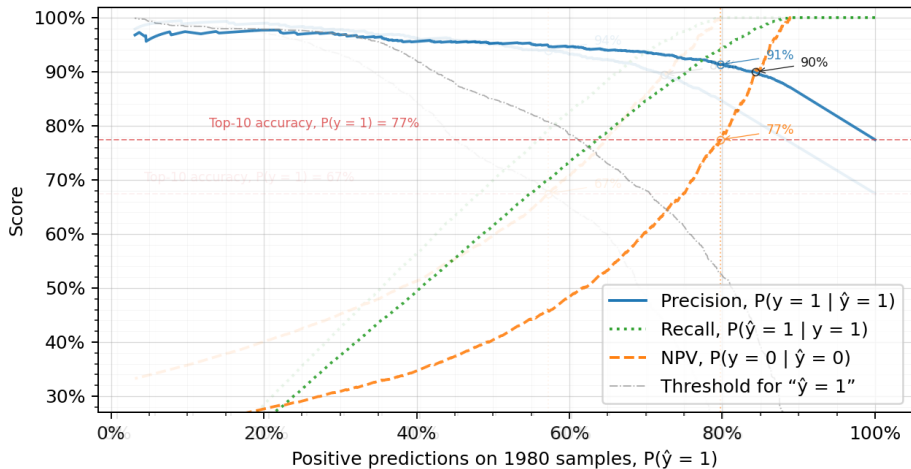
$0 \leq \text{is_sum_ok} \leq 1$



$13 \leq \text{char_len} \leq 79$



$129 \leq \text{nmr_data_char_len} \leq 919$



thx.

References I

- [1] Marvin Alberts, Oliver Schilter, Federico Zipoli, Nina Hartrampf, and Teodoro Laino. “Unraveling Molecular Structure: A Multimodal Spectroscopic Dataset for Chemistry”. In: *NeurIPS 2024 Datasets and Benchmarks Track*. 2024. [url].
- [2] Roberto Cipolla, Yarin Gal, and Alex Kendall. “Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics”. In: *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2018). [url].

